CSE 270 Sports Analytics

Basketball Individual performance

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NBA Individual performance

NBA Individual performance

2 main stages of the game.

- Attacking
- Defending

https://www.youtube.com/watch?v=wYjp2zoqQrs

NBA Individual performance

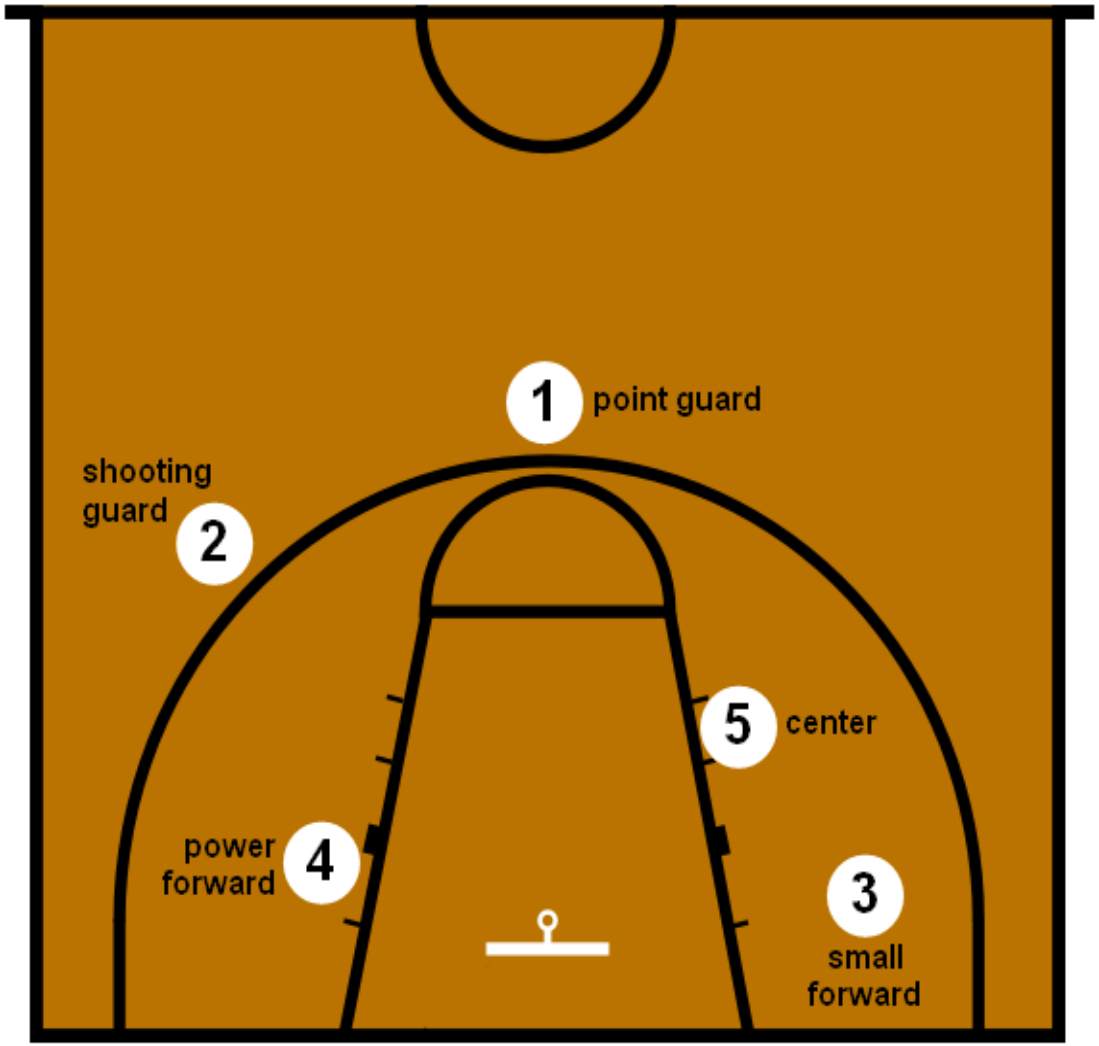
5 main positions on the field.

- Center - On offense, the center tries to score on close shots and rebound. But on defense, the center tries to block opponents' shots and rebound their misses.
- Power Forward - The power forward does many of the things a center does, playing near the basket while rebounding and defending taller players. But power forwards also take longer shots than centers.
- Small Forward - The small forward plays against small and large players. They roam all over on the court. Small forwards can score from long shots and close ones.
- Point Guard - The point guard runs the offense.The point guard defends the opponent's point guard and tries to steal the ball.
- Shooting Guard - The shooting guard can make shots from long distance and also is a good dribbler.

Let's watch a short video,

https://www.youtube.com/watch?v=4_4CymXARWQ

NBA Individual performance



NBA Individual performance

Defining attributes for each position.

- Center - The center is the tallest player on each team, playing near the basket.
- Power Forward - The power forward is very close to center player, but in this position they should also have good shooting attributes.
- Small Forward - The small forward should have good shooting efficiency in close and far distances.
- Point Guard - The point guard should have good defending characteristics.
- Shooting Guard - The shooting guard should have good dribbling attributes and shooting accuracy in long ranges.

What measures other than the height and shooting can be used to position the players on the field ?

NBA Individual performance

In game statistics

- STEALS
- TURNOVERS - Turnover occurs when a team loses possession of the ball to the opposing team before a player takes a shot at their team's basket.
  https://www.youtube.com/shorts/I4MDXOWrUmk
- REBOUNDS
  1. OFFENSIVE REBOUNDS - An offensive rebound occurs when offensive players regain possession of the ball after a missed shot.
  2. DEFENSIVE REBOUNDS - Defensive rebounds occur when defensive players gain possession of the ball after an offensive player misses the shot.
- BLOCKS - Block occurs when a defensive player legally deflects a field goal attempt from an offensive player to prevent a score.
- FOULS
  1. PERSONAL FOULS
  2. FLAGRANT FOULS
  3. TECHINCAL FOULS

NBA Individual performance

Dataset

```
str(nba_players)
```

```
## 'data.frame':    9708 obs. of  53 variables:
##  $ X     : int  14967 14968 14969 14970 14971 14972 14973 14974 14975 14976 ...
##  $ Year  : int  2001 2001 2001 2001 2001 2001 2001 2001 2001 2001 ...
##  $ Player: chr  "Mahmoud Abdul-Rauf" "Tariq Abdul-Wahad" "Shareef Abdur-Rahim" "Cory Alexander" ...
##  $ Pos   : chr  "PG" "SG" "SF" "PG" ...
##  $ Age   : int  31 26 24 27 23 23 23 25 24 30 ...
##  $ Tm    : chr  "VAN" "DEN" "VAN" "ORL" ...
##  $ G     : int  41 29 81 26 65 38 27 82 37 82 ...
##  $ GS    : int  0 12 81 0 24 6 18 82 2 36 ...
##  $ MP    : int  486 420 3241 227 1382 472 910 3129 288 1710 ...
##  $ PER   : num  16.7 5.8 19.1 6.3 11.6 6.8 14.2 22.9 12.1 8.7 ...
##  $ TS.   : num  0.514 0.438 0.549 0.407 0.484 0.404 0.519 0.61 0.429 0.455 ...
##  $ X3PAr : num  0.057 0.09 0.05 0.286 0.08 0.056 0.091 0.357 0.357 0.012 ...
##  $ FTr   : num  0.118 0.324 0.415 0.321 0.262 0.253 0.266 0.299 0.155 0.471 ...
##  $ ORB.  : num  1.2 3.6 6.2 0 3.5 4.9 2.8 3.7 1.6 4.9 ...
##  $ DRB.  : num  4.7 11.8 19.6 12.5 8.4 9.9 7.6 11.8 10.6 12.7 ...
##  $ TRB.  : num  2.9 7.6 13 6.1 5.9 7.5 5.1 7.8 6.1 8.7 ...
##  $ AST.  : num  32.7 8.1 14.3 23.5 8 6.9 8.6 20.4 34.6 7 ...
##  $ STL.  : num  1 1.7 1.5 3.6 1.7 1.7 1.7 2.1 2.4 0.8 ...
##  $ BLK.  : num  0.1 2.1 1.6 0 0.3 0.5 0.2 0.5 0 1.2 ...
##  $ TOV.  : num  9.1 21.1 13.2 28.1 10.5 9.6 10.9 12.1 18.2 14.8 ...
##  $ USG.  : num  26.8 17.1 24.6 17.3 23.2 21.1 24.3 24.4 17.2 21.7 ...
##  $ blanl : logi  NA NA NA NA NA NA ...
##  $ OWS   : num  0.7 -0.7 5.6 -0.5 0.3 -0.5 0.9 11.5 0.1 -1.3 ...
##  $ DWS   : num  0 0.3 2.9 0.3 0.5 0.4 0.1 2.2 0.2 1.4 ...
##  $ WS    : num  0.7 -0.4 8.5 -0.1 0.8 -0.1 0.9 13.7 0.3 0 ...
##  $ WS.48 : num  0.07 -0.049 0.126 -0.031 0.027 -0.013 0.048 0.211 0.055 0.001 ...
##  $ blank2: logi  NA NA NA NA NA NA ...
##  $ OBPM  : num  0.8 -5.9 1.3 -5.9 -1.8 -4.8 -0.3 6.5 -1.8 -5.1 ...
##  $ DBPM  : num  -5.3 0 0.9 -0.1 -2.5 -1.8 -2.9 -0.9 -2.7 -1.8 ...
##  $ BPM   : num  -4.5 -5.9 2.2 -6.1 -4.4 -6.6 -3.2 5.6 -4.5 -6.9 ...
##  $ VORP  : num  -0.3 -0.4 3.5 -0.2 -0.8 -0.5 -0.3 6 -0.2 -2.1 ...
##  $ FG    : int  120 43 604 18 239 62 177 628 30 237
```

NBA Individual performance

Let's select the performance metrics.

```
nba_players <- nba_players %>%
  select(Year, Player, Pos,
         G, MP, STL, TOV, ORB, DRB, PF, BLK, PTS, AST,
         FT, FTA, FG, FGA, X3P, X3PA, ) %>%
  arrange(Player, Year)
head(nba_players, 3)
```

```
##   Year       Player Pos  G   MP STL TOV ORB DRB  PF BLK PTS AST FT FTA  FG FGA
## 1 2001  A.C. Green  PF 82 1411  30  45 107 206 119   8 367  39 79 111 144 324
## 2 2001 A.J. Guyton  PG 33  630   9  24  10  26  35   5 198  64 15  18  78 192
## 3 2002 A.J. Guyton  PG 45  607  10  37  12  32  23   7 244  81 22  27  88 244
##   X3P X3PA
## 1   0    6
## 2  27   69
## 3  46  123
```

NBA Individual performance

Let's try to evaluate the player's performance based on the amount of time they spent on the field.
We will use the players' historical data to measure their performance for each Year, by taking 5 preceding Years only in order to avoid one Year stars. We will use success/failure approach by dividing the metrics of the players into successful and non-succesfull actions and then compare their success rate with their playing Minutes.

NBA Individual performance

Let's start with defending attributes.

Successful actions, when defending

- STEALS
- BLOCKS
- DEFENSIVE REBOUNDS

Non Successful actions, when defending

- PERSONAL FOULS
- TURNOVER

NBA Individual performance

Let's calculate the difference between the cumulative success actions and cumulative failure actions.

```
nba_players_defense <- nba_players %>%
  arrange(Player, Year)  %>%
  group_by(Player) %>%
  mutate(CUM_STL = rollmean(STL, k = 5, fill = NA, align="right"),
         CUM_TOV = rollmean(TOV, k = 5, fill = NA, align="right"),
         CUM_DRB = rollmean(DRB, k = 5, fill = NA, align="right"),
         CUM_PF = rollmean(PF, k = 5, fill = NA, align="right"),
         CUM_BLK = rollmean(BLK, k = 5, fill = NA, align="right"),
         CUM_MP = rollmean(MP, k = 5, fill = NA, align = "right"),
         CUM_SUCCESS_1 = CUM_STL - CUM_TOV,
         CUM_SUCCESS_2 = CUM_STL - CUM_PF,
         CUM_SUCCESS_3 = CUM_DRB - CUM_TOV,
         CUM_SUCCESS_4 = CUM_DRB - CUM_PF,
         CUM_SUCCESS_5 = CUM_BLK - CUM_TOV,
         CUM_SUCCESS_6 = CUM_BLK - CUM_PF,
         SUCCESS_1 = STL - TOV,
         SUCCESS_2 = STL - PF,
         SUCCESS_3 = DRB - TOV,
         SUCCESS_4 = DRB - PF,
         SUCCESS_5 = BLK - TOV,
         SUCCESS_6 = BLK - PF,
         ) %>%
  drop_na() %>%
  ungroup()
```

NBA Individual performance

Let's check if the cumulative metrics are correlated to the amount of time the players spent on the field.

```
cor_with_MP <- function(x){
  return(cor(x, nba_players_defense$MP))
}

nba_players_defense %>%
  select(CUM_SUCCESS_1, CUM_SUCCESS_2, CUM_SUCCESS_3,
         CUM_SUCCESS_4, CUM_SUCCESS_5, CUM_SUCCESS_6, CUM_MP) %>%
  apply(2,cor_with_MP)
```

```
## CUM_SUCCESS_1 CUM_SUCCESS_2 CUM_SUCCESS_3 CUM_SUCCESS_4 CUM_SUCCESS_5
##    -0.5037294    -0.3629738     0.4011041     0.4643897    -0.4968245
## CUM_SUCCESS_6        CUM_MP
##    -0.5414721     0.7740168
```

NBA Individual performance

Now let's do the same for Seasonal metrics.

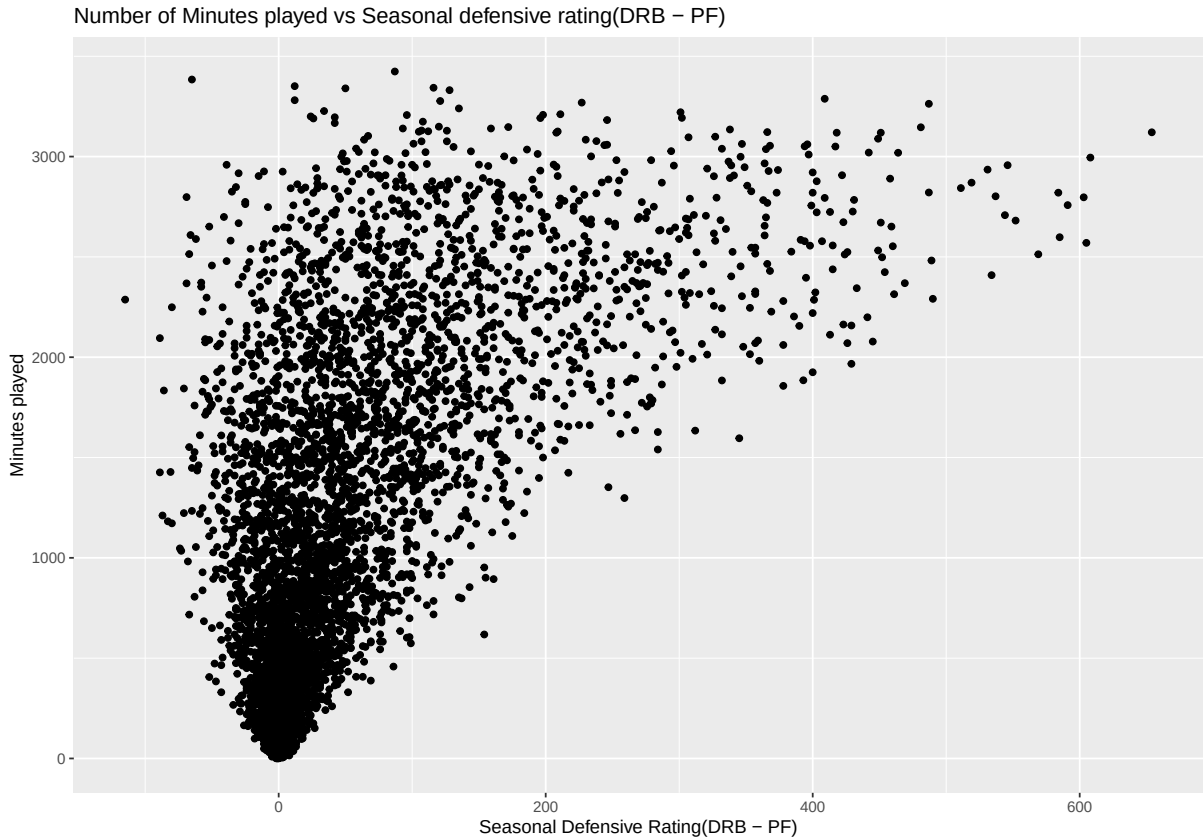
```
nba_players_defense %>%
  select(SUCCESS_1, SUCCESS_2, SUCCESS_3,
         SUCCESS_4, SUCCESS_5, SUCCESS_6) %>%
  apply(2,cor_with_MP)
```

```
##  SUCCESS_1  SUCCESS_2  SUCCESS_3  SUCCESS_4  SUCCESS_5  SUCCESS_6
## -0.6568455 -0.6323585  0.6012873  0.6033670 -0.6796871 -0.7956140
```

NBA Individual performance

The metric with the highest positive correlation rate is the Seasonal difference between the cumulative number of defensive rebounds made and cumulative number of turnovers made.

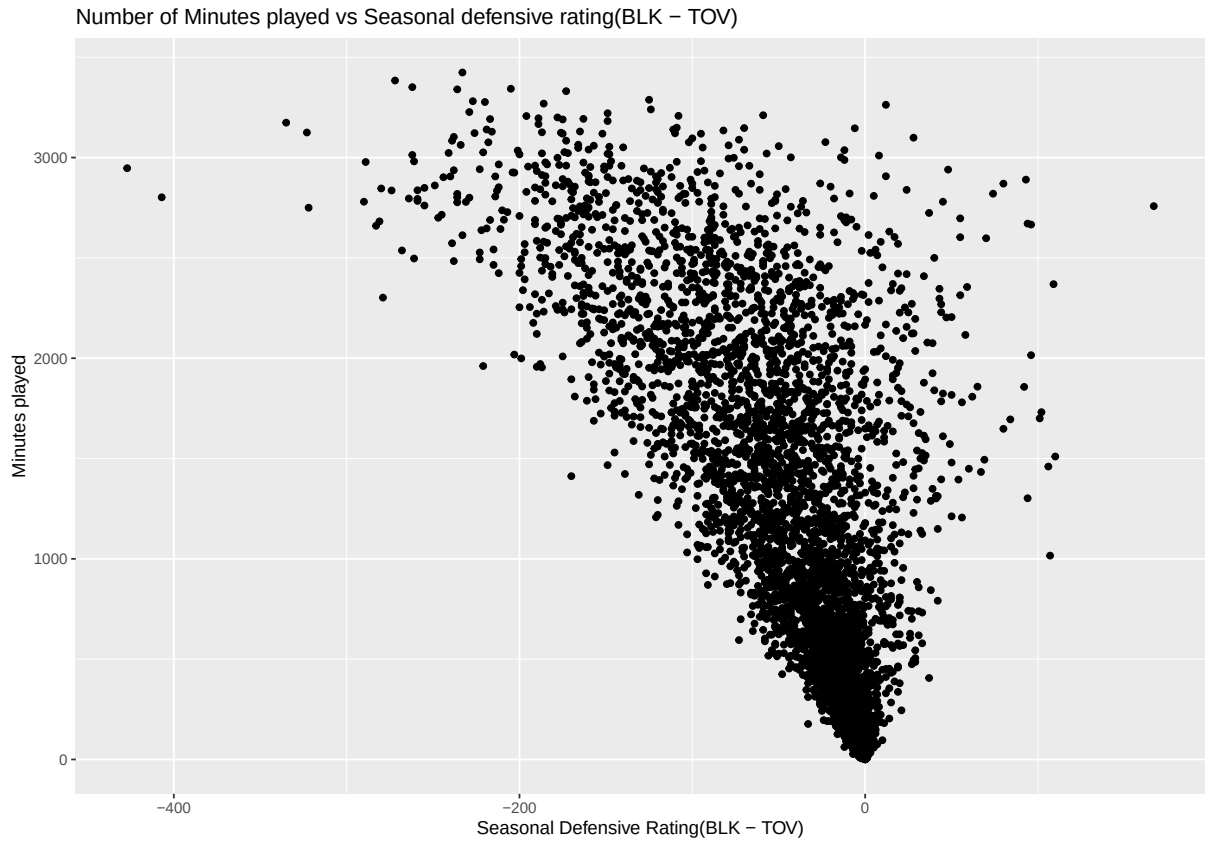
```
ggplot(nba_players_defense, aes(x = SUCCESS_4, y = (MP))) +
  geom_point() +
  ggtitle("Number of Minutes played vs Seasonal defensive rating(DRB - PF)") +
  xlab("Seasonal Defensive Rating(DRB - PF)") +
  ylab("Minutes played")
```



NBA Individual performance

The metric with the lowest positive correlation rate is the difference between the cumulative number of blocks made and cumulative number of turnovers made.

```
ggplot(nba_players_defense, aes(x = SUCCESS_5, y = (MP))) +
  geom_point() +
  ggtitle("Number of Minutes played vs Seasonal defensive rating(BLK - TOV)") +
  xlab("Seasonal Defensive Rating(BLK - TOV)") +
  ylab("Minutes played")
```



NBA Individual performance

Let's drop the difference between number of steals made and fouls made, as it has almost 0 correlation with the number of Minutes played and then fit a model with the defensive power ratings.

```
nba_players_defense <- nba_players_defense %>%
  select(-CUM_SUCCESS_2, SUCCESS_2)

summary(lm(MP ~ (SUCCESS_4 + CUM_SUCCESS_4 + CUM_SUCCESS_5
                  + SUCCESS_5 + CUM_MP ), data = nba_players_defense))
```

```
##
## Call:
## lm(formula = MP ~ (SUCCESS_4 + CUM_SUCCESS_4 + CUM_SUCCESS_5 +
##     SUCCESS_5 + CUM_MP), data = nba_players_defense)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2499.24  -196.48   -19.07   185.62  1861.77
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -18.0697    11.0865   -1.63    0.103
## SUCCESS_4       4.5796     0.1115   41.06   <2e-16 ***
## CUM_SUCCESS_4  -4.2496     0.1287  -33.03   <2e-16 ***
## CUM_SUCCESS_5   9.8201     0.2106   46.62   <2e-16 ***
## SUCCESS_5     -11.1486     0.1798  -61.99   <2e-16 ***
## CUM_MP          0.8950     0.0122   73.36   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 336 on 4753 degrees of freedom
## Multiple R-squared:  0.8532, Adjusted R-squared:  0.8531
## F-statistic:  5526 on 5 and 4753 DF,  p-value: < 2.2e-16
```

NBA Indvidual performance.

Dropping the cumulative average of Minutes played to understand the model's power on estimating the players' number of Minutes on the field without preceding 5 Year.

```
summary(lm(MP ~ (SUCCESS_4 + CUM_SUCCESS_4 + CUM_SUCCESS_5
                 + SUCCESS_5 ), data = nba_players_defense))
```

```
##
## Call:
## lm(formula = MP ~ (SUCCESS_4 + CUM_SUCCESS_4 + CUM_SUCCESS_5 +
##     SUCCESS_5), data = nba_players_defense)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4094.6  -361.4   -87.1   289.1  2350.4
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)   587.7982    10.7979  54.436  < 2e-16 ***
## SUCCESS_4       4.8541     0.1627  29.827  < 2e-16 ***
## CUM_SUCCESS_4  -0.5806     0.1731  -3.354 0.000802 ***
## CUM_SUCCESS_5   2.7103     0.2730   9.927  < 2e-16 ***
## SUCCESS_5     -11.3476     0.2625 -43.222  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 490.6 on 4754 degrees of freedom
## Multiple R-squared:  0.687,  Adjusted R-squared:  0.6868
## F-statistic:  2609 on 4 and 4754 DF,  p-value: < 2.2e-16
```

NBA Individual performance

We can see that all the cumulative metrics of the player's except the average number of Minutes played in preceeding 5 seaons, do not have as much significance as Seasonal values, so we can conclude that the number of Minutes the players will have on the field is connected to their previous Year statistics, but can change based on their Seasonal performance.

NBA Individual performance

For the attacking attributes we will calculate the difference between the attempts made for successful actions and actual successful actions.

Successful actions, when attacking

- FIELD GOALS
- 3 POINT GOALS
- FREE THROWS
- OFFENSIVE REBOUNDS
- POINTS SCORED
- ASSISTS MADE

Non Successful actions, when attacking

- FIELD GOALS ATTEMPTS
- 3 POINT ATTEMPTS
- FREE THROW ATTEMPTS

NBA Individual performance

```
nba_players_offense <- nba_players %>%
  arrange(Player, Year)  %>%
  group_by(Player) %>%
  mutate(CUM_FG = rollmean(FG, k = 5, fill = NA, align="right"),
         CUM_FGA = rollmean(FGA, k = 5, fill = NA, align="right"),
         CUM_X3P = rollmean(X3P, k = 5, fill = NA, align="right"),
         CUM_X3PA = rollmean(X3PA, k = 5, fill = NA, align="right"),
         CUM_FT = rollmean(FT, k = 5, fill = NA, align="right"),
         CUM_FTA = rollmean(FTA, k = 5, fill = NA, align="right"),
         CUM_AST = rollmean(AST, k = 5, fill = NA, align="right"),
         CUM_PTS = rollmean(PTS, k = 5, fill = NA, align="right"),
         CUM_ORB = rollmean(PTS, k = 5, fill = NA, align="right"),
         CUM_MP = rollmean(MP, k = 5, fill = NA, align="right"),
         SUCCESS_1 = FGA - FG,
         SUCCESS_2 = X3PA - X3P,
         SUCCESS_3 = FTA - FT,
         CUM_SUCCESS_1 = CUM_FGA - CUM_FG,
         CUM_SUCCESS_2 = CUM_X3PA - CUM_X3P,
         CUM_SUCCESS_3 = CUM_FTA - CUM_FT,
         ) %>%
  drop_na() %>%
  ungroup()
```

NBA Individual performance

Let's check if the cumulative metrics are correlated to the amount of time the players spent on the field.

```
cor_with_MP <- function(x){
  return(cor(x, nba_players_offense$MP))
}

nba_players_offense %>%
  select(CUM_SUCCESS_1, CUM_SUCCESS_2, CUM_SUCCESS_3, CUM_AST, CUM_MP, CUM_PTS) %>%
  apply(2,cor_with_MP)
```

```
## CUM_SUCCESS_1 CUM_SUCCESS_2 CUM_SUCCESS_3       CUM_AST        CUM_MP
##     0.7102738     0.4451066     0.5211743     0.5399877     0.7740168
##       CUM_PTS
##     0.7266231
```

NBA Individual performance

Let's check if the Seasonal metrics are correlated to the amount of time the players spent on the field.

```
cor_with_MP <- function(x){
  return(cor(x, nba_players_offense$MP))
}

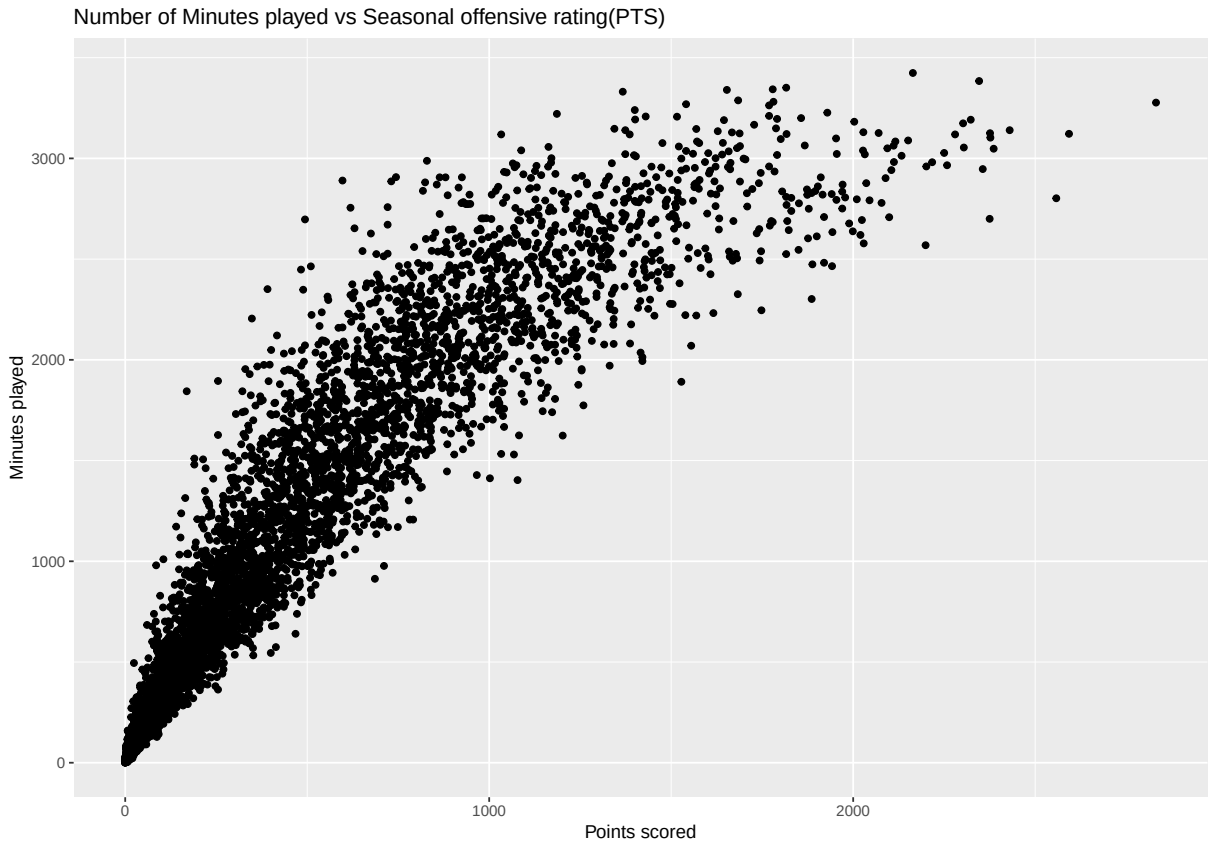
nba_players_offense %>%
  select(SUCCESS_1, SUCCESS_2, SUCCESS_3, AST, PTS) %>%
  apply(2,cor_with_MP)
```

```
## SUCCESS_1 SUCCESS_2 SUCCESS_3       AST       PTS
## 0.9168942 0.6483738 0.6840179 0.7301981 0.9194013
```

NBA Individual performance

The Seasonal metric with the highest correlation is the difference between the number of points scored by the player.

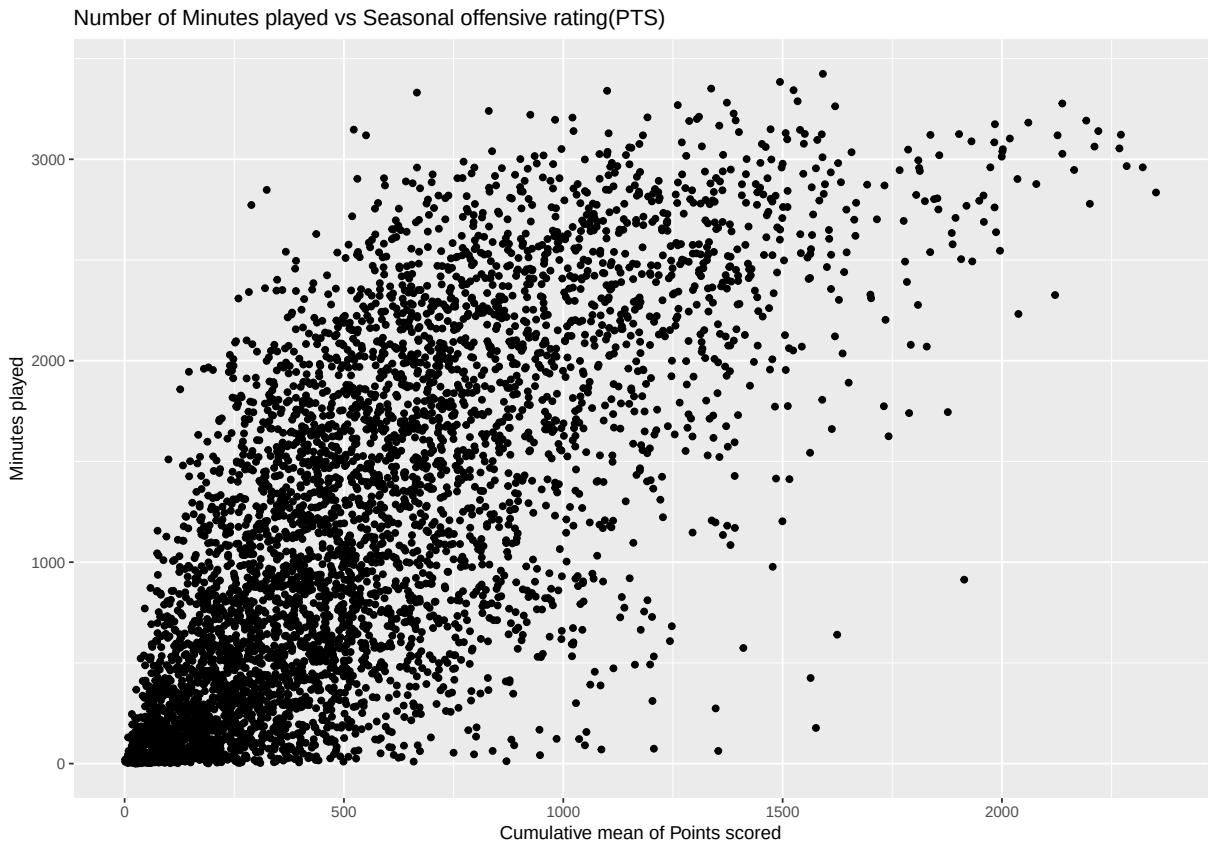
```
ggplot(nba_players_offense,aes(x = PTS, y = MP)) +
  geom_point() +
  ggtitle("Number of Minutes played vs Seasonal offensive rating(PTS)") +
  xlab("Points scored") +
  ylab("Minutes played")
```



NBA Individual performance

The cumulative metric with the highest correlation is again the number of points scored.

```
ggplot(nba_players_offense,aes(x = CUM_PTS, y = MP)) +
  geom_point() +
  ggtitle("Number of Minutes played vs Seasonal offensive rating(PTS)") +
  xlab("Cumulative mean of Points scored") +
  ylab("Minutes played")
```



NBA Individual performance

```
summary(lm(MP ~ (PTS + AST + SUCCESS_1 +
                  + CUM_SUCCESS_1 + CUM_PTS + CUM_AST + CUM_MP ), data = nba_players_offense))
```

```
##
## Call:
## lm(formula = MP ~ (PTS + AST + SUCCESS_1 + +CUM_SUCCESS_1 + CUM_PTS +
##     CUM_AST + CUM_MP), data = nba_players_offense)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1258.23  -107.74    -4.02   108.83   884.44
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -19.52353    6.62001  -2.949   0.0032 **
## PTS             1.02572    0.04247  24.150   <2e-16 ***
## AST             1.13187    0.07087  15.971   <2e-16 ***
## SUCCESS_1       1.80142    0.10054  17.917   <2e-16 ***
## CUM_SUCCESS_1  -1.32269    0.11525 -11.476   <2e-16 ***
## CUM_PTS        -1.09525    0.04974 -22.021   <2e-16 ***
## CUM_AST        -1.07336    0.07228 -14.850   <2e-16 ***
## CUM_MP          0.93532    0.01043  89.656   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 195.6 on 4751 degrees of freedom
## Multiple R-squared:  0.9503, Adjusted R-squared:  0.9502
## F-statistic: 1.297e+04 on 7 and 4751 DF,  p-value: < 2.2e-16
```

NBA Indvidual performance.

Dropping the cumulative average of Minutes played to understand the model's power on estimating the players' number of Minutes on the field without preceding 5 Years.

```
summary(lm(MP ~ (PTS + AST + SUCCESS_1 +
                  + CUM_SUCCESS_1 + CUM_PTS + CUM_AST), data = nba_players_offense))
```

```
##
## Call:
## lm(formula = MP ~ (PTS + AST + SUCCESS_1 + +CUM_SUCCESS_1 + CUM_PTS +
##     CUM_AST), data = nba_players_offense)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2330.60  -209.90   -47.15   172.22  1569.45
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)   361.00290    8.33454  43.314  < 2e-16 ***
## PTS             0.91605    0.06965  13.152  < 2e-16 ***
## AST             1.07135    0.11626   9.215  < 2e-16 ***
## SUCCESS_1       1.98581    0.16491  12.042  < 2e-16 ***
## CUM_SUCCESS_1  -0.60034    0.18861  -3.183 0.001467 **
## CUM_PTS         0.03366    0.07894   0.426 0.669840
## CUM_AST        -0.41722    0.11797  -3.537 0.000409 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 320.9 on 4752 degrees of freedom
## Multiple R-squared:  0.8661, Adjusted R-squared:  0.866
## F-statistic:  5125 on 6 and 4752 DF,  p-value: < 2.2e-16
```

NBA Individual performance

We can again see that the Seasonal metrics have higher contribution to the number of Minutes played, and we can conclude that the players who are making more attempts to score field goal are generally playing more.